DO NOW — RECALL

Karnaugh maps and simplification

Name: _____ Class: _____ Date: _____

1. Simplify ¬(A ∨ B).

2. Keys 7 and 12 hash to index 2 in a table. What is that called?

3. What is 1 XOR 1?

4. If A changes from 0 to 1 while B is always 1, which variable is constant?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

¬A ∧ ¬B.

2. Model answer

A collision.

3. Model answer

0.

4. Model answer

B, at value 1.

Vocabulary — know these terms

Karnaugh map — Grid for simplifying Boolean functions using adjacent combinations.

Gray code — Ordering in which adjacent binary labels differ by one bit.

group — A rectangle containing 1, 2, 4, 8, … adjacent 1-cells.

constant variable — A variable with the same value in all cells of a group.

wraparound — The opposite edges of a Karnaugh map are adjacent.

NEW LEARNING

Read, annotate and explain

Read the labels

Rows represent A, and columns use Gray order BC: 00, 01, 11, 10.

Group neighbouring ones

The four 1s form one 2×2 rectangle. Adjacency across rows is allowed.

Keep what is constant

A and C change through the group, but B remains 0. So the simplified result is ¬B.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Four 1s fill the entire A=0 row. Simplify the output.

Predict and explain

B and C each vary across the row; neither is needed in the expression.

Check and extend

A stays 0 in the whole group. The simplified term is ¬A.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Why do we group neighbouring 1s in a Karnaugh map?

Question 2

The whole A=1 row contains four 1s. Which simplified term remains?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Columns BC=00 and BC=10 are joined by wraparound adjacency. Four 1s occupy both these columns. Simplify.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A group of four 1s can remove two changing variables.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.